By building components and layouts, you better understand how D3 works, but there's another reason why they're so valuable: reusability. You've built a chart using a layout and component (no matter how simple) that you wrote yourself. You could use either in tandem with another layout or component, or on its own, with any data visualization charts you use elsewhere.

Infoviz term: reusable charts

After you've worked with components, layouts, and controls in D3, you may start to wonder if there's a higher level of abstraction available that could combine layouts and controls in a reusable fashion. That level of abstraction has been referred to as a *chart*, and the creation of reusable charts has been of great interest to the D3 community.

This has led to the development of several APIs on top of D3, such as NVD3, D4 (for generic charts), and my own `d3.carto.map` (for web mapping, not surprisingly). It's also led The Miso Project to develop `d3.chart`, a framework for reusable charts. If you're interested in using or developing reusable charts, you may want to check these out:

d3.chart http://misoproject.com/d3-chart/

d3.carto.map https://github.com/emeeks/d3-carto-map

http://visible.io (D4)

http://nvd3.org (NVD3)

You may also try your hand at building more responsive components that automatically update when you call them again, like the axis and brushes we dealt with in the last chapter. Or you may try creating controls like `d3.brush` and behaviors like `d3.behavior.drag`. Regardless of how extensively you follow this pattern, I recommend that you look for instances when your information visualization can be abstracted into layouts and components, and try to create those instead of building another one-off visualization. By doing that, you'll develop a higher level of skill with D3 and fill your toolbox with your own pieces for later work.

10.3 Summary

This chapter showed you how to follow two of the patterns that appear in D3: layouts and components. You learned how to create both of these in a way that you can reuse them and combine them in a single chart:

- Create a general layout structure with getter and setter functions.
- Build the functionality necessary for the layout to modify sent data with attributes for drawing.
- Make a layout dynamically change size based on a size setting.
- Modify sent data to dynamically change the size of individual grid cells.